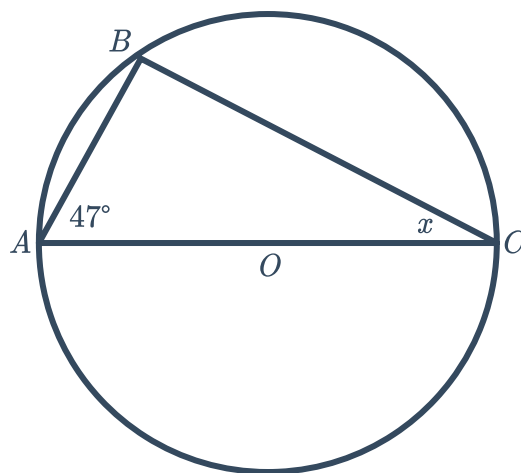


Missing angles



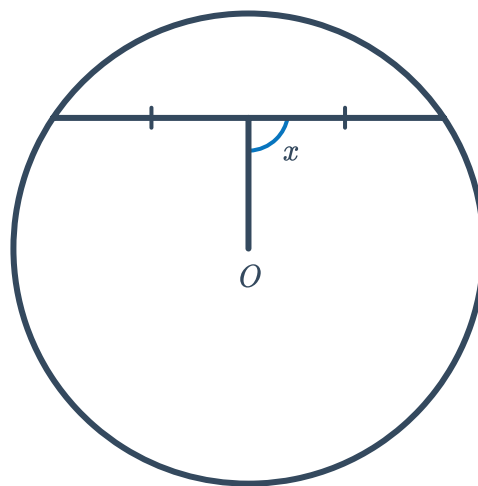
- 1 In the given diagram, AC is a straight line passing through centre O :

Solve for x .



- 2 Consider the circle with centre O shown:

Solve for x .

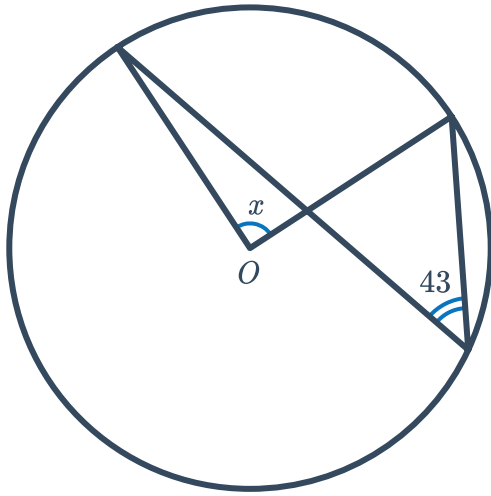




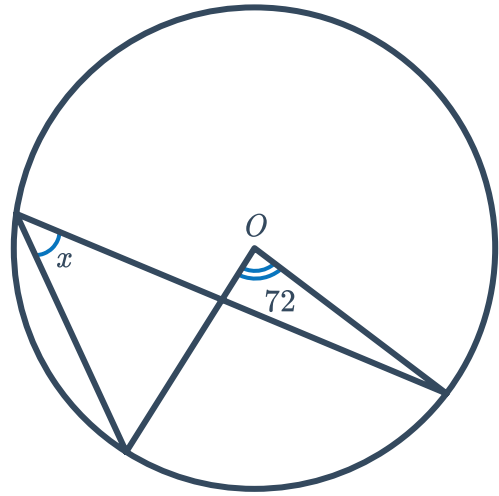
Missing angles (Extended)

3 For each of the following circles with centre O , solve for x :

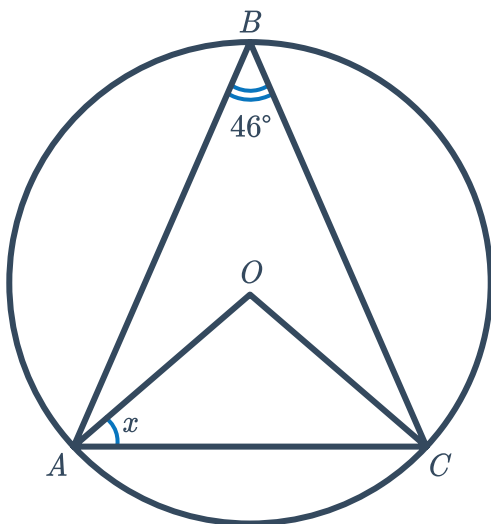
a



b



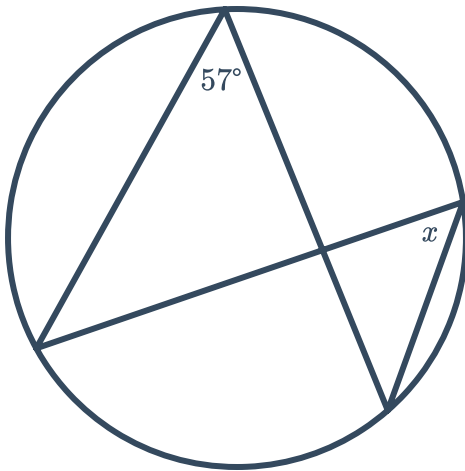
c



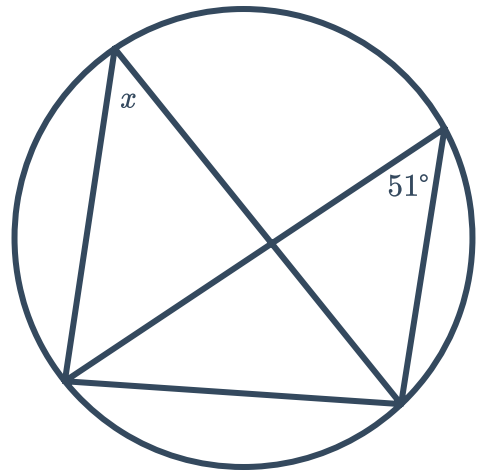
4 For each of the following circles, solve for



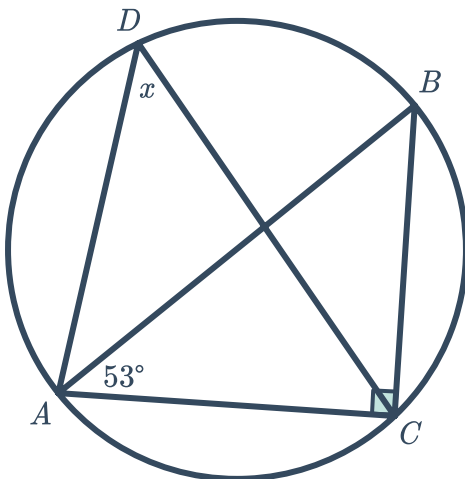
a



b

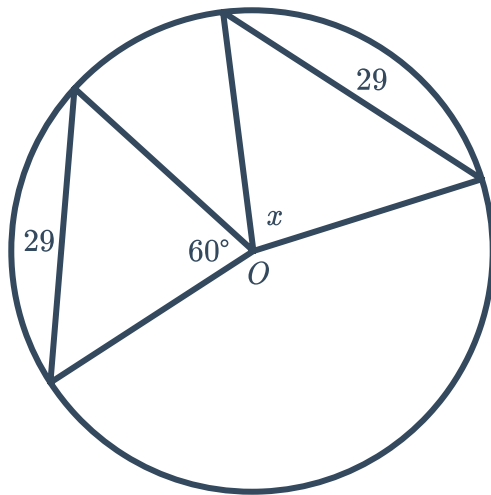


c

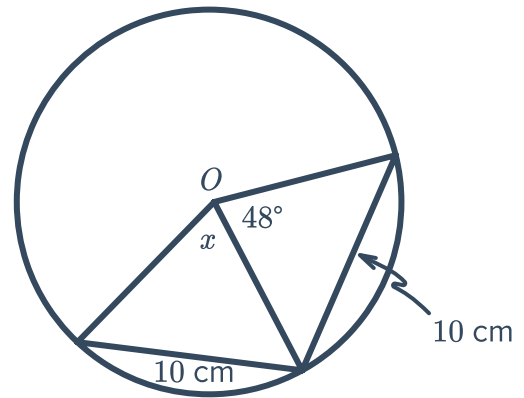


5 For each of the following circles with centre  solve for x :

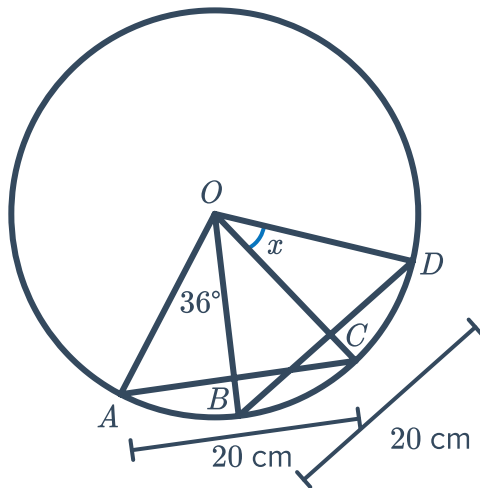
a



b



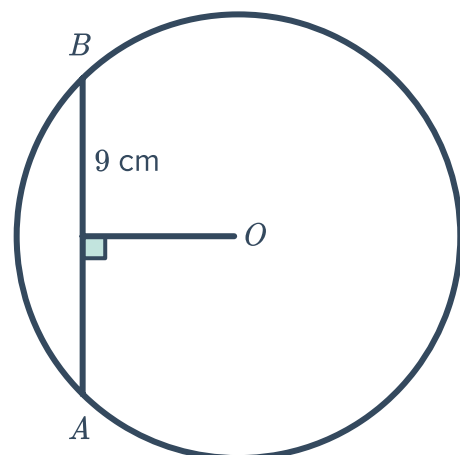
c



Missing lengths

6 Consider the following circle where O is the centre:

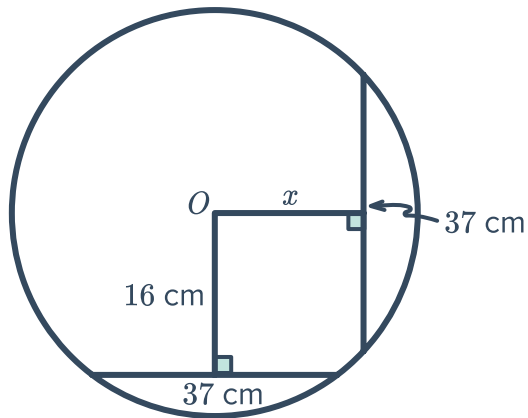
Find the length AB .



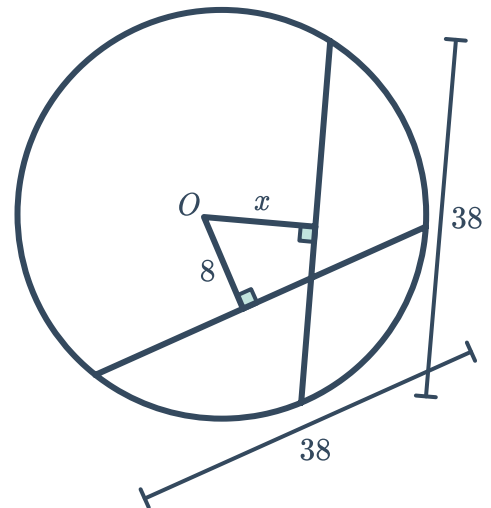


7 For each of the following circles with centre O , solve for x :

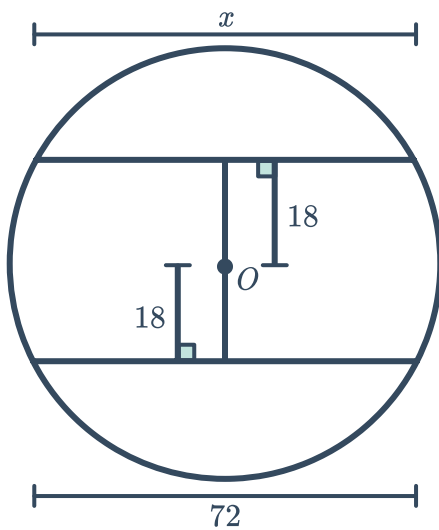
a



b

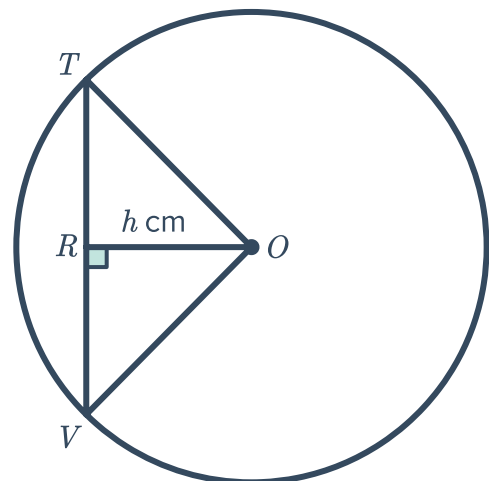


c



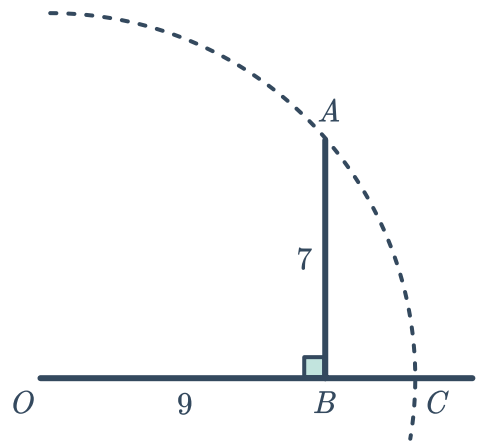
8 The given circle has $TR = 12$ cm, the radius is 13 cm, and OR has length h cm.

Find the area of $\triangle TOV$.



- 9 The diagram shows AC as the arc of a circle with O as its centre, $OB = 9$, and $AB = 7$:

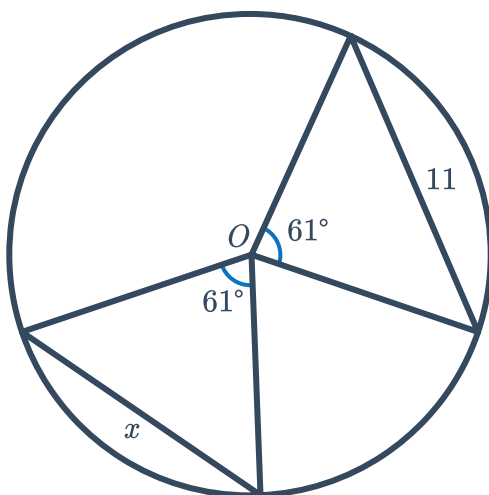
- a Find the exact length of OA .
b Hence, find the exact length of BC .



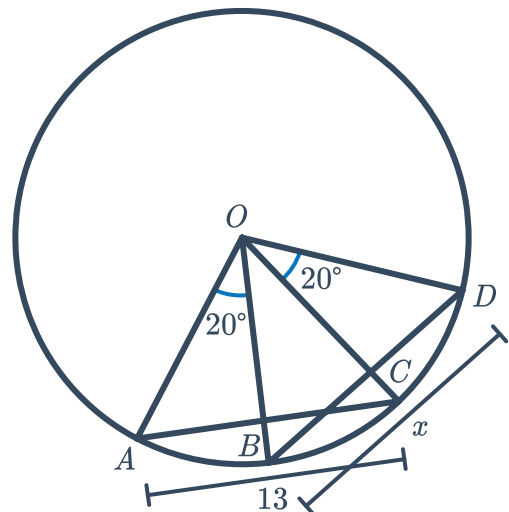
Missing lengths (Extended)

- 10 For each of the following circles with centre O , solve for x :

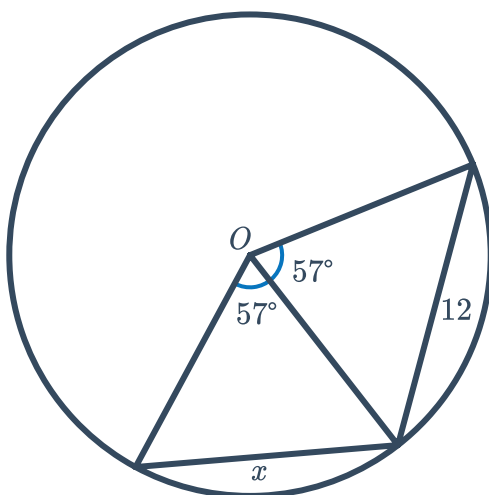
a



b

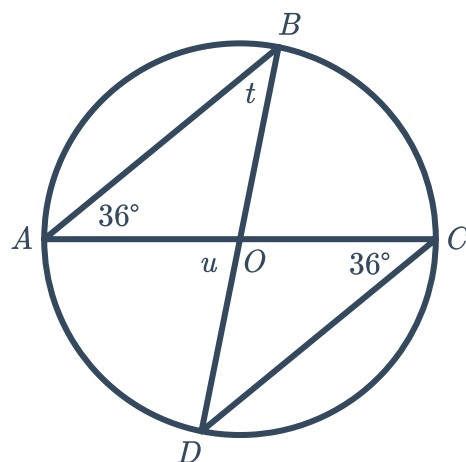


c



- 11 Consider the following circle where O is the centre:

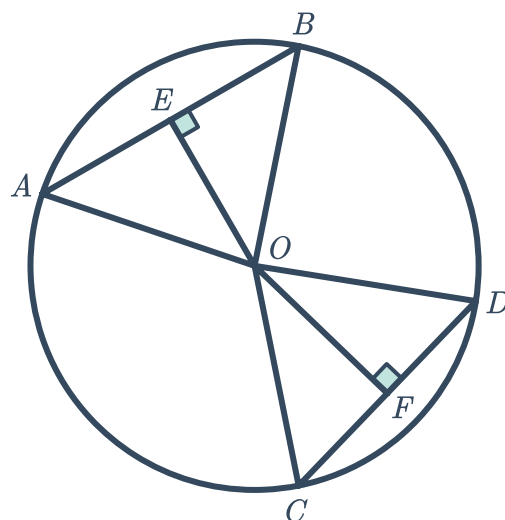
- a Solve for t .
- b Solve for u .



Proofs

- 12 Let O be the centre of the given circle with $AB = CD$.

- a Prove that $\triangle ABO$ and $\triangle CDO$ are congruent.
- b Hence, prove that $\triangle BEO$ and $\triangle DFO$ are congruent.



- 13 (Extended)

Let O be the centre of the given circle:



Prove that x and y are complementary angles.

